Reg.No : ……

Name : …….

MAHATHMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2026

FIFTH SEMESTER

DISCIPLINE SPECIFIC CORE COURSE (DSC A) – MG5DSCBTG300- RECOMBINANT DNA TECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1½ **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part A

One word Answer

Answer any **10** questions

Each question carries **2** marks

1. Name the enzyme that joins two DNA fragments by forming phosphodiester bonds. *[K] / [CO1]*
2. Identify the sequencing platform based on semiconductor detection. *[U] / [CO3]*
3. Name the method that uses high-velocity particles for DNA delivery *[K] / [CO2]*
4. Which marker detects single-base variations? [K]/ [CO3]
5. Name one selectable marker used in plant genetic transformation. *[K] / [CO2]*
6. What does PCR stand for? *[U] / [CO3]*
7. Which reporter protein exhibits green fluorescence? *[K] / [CO2]*
8. Name one E. coli cloning vector derived from pBR322. *[K] / [CO1]*
9. What is the term for short double-stranded oligonucleotides used in DNA-end modification? *[U] / [CO1]*
10. Write one application of chromosome walking. *[A] /[CO3]*
11. Identify the selectable marker that confers resistance to hygromycin. *[U] / [CO2]*
12. Which technique detects protein–DNA interaction by retardation of DNA migration? *[K] / [CO3]*

(2×10= 20)

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

13. Explain the different enzymes used for in vitro modification of nucleic acids, with special reference to restriction endonucleases and DNA ligases. [U]/[CO1]
14. Discuss the desirable properties of a plasmid cloning vector with suitable examples. [K]/ [CO1]
15. Explain the principle and workflow of Illumina sequencing [K]/ [CO3]
16. Describe insertional inactivation as a method for screening recombinant clones. [U]/ [CO2]
17. Describe electroporation as a method of gene transfer. Mention its advantages and limitations. [K]/ [CO2]
18. Compare M13 vectors and λ phage vectors with respect to their important features and applications. [An]/[CO1]

(5×4= 20)

Part C

Essay Questions

Answer any **1** questions

Each questions carries **10** marks

19. Illustrate gene therapy as an application of recombinant DNA technology. Explain its basic concept, approaches and potential applications [An]/[CO3]
20. Describe Agrobacterium-mediated transformation in plants in detail. [U]/ [CO2]

(10×1= 10)